

Question number	Answer	Additional guidance	Mark
4(a)(i)	A description that includes the following points: • general trend the higher the (resting) heart rate the lower the life expectancy/ no clear relationship between the (resting) heart rate the life expectancy (1) • the higher the (resting) heart rate the lower the (mean) mass (1) • the higher the (resting) heart rate the lower the (mean resting) metabolic rate (1)	Statements can be either way round accept negative correlation accept converse accept converse	Expert (2)

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Question number	Answer	Additional guidance	Mark
4(a)(ii)	A calculation with the following steps • calculation of oxygen used per kilogram(1) • calculation of oxygen used for 4500kg elephant(1) • correct answer in standard form (1)	$(1400 \div 5780) = 0.24$ $(4500 \times 0.24) = 1080$ $1.09 \times 10^3 / 1.0896 \times 10^3$ ignoring any units ECF for mps 2 and 3 alternate method working out: 1 mark for $1400 \times 4500 = 6\,300\,000$ for 2 marks $(4500 \times 1400) \div 5780 = 1089.96$	Expert (3)

1 mark	2 marks	3 marks
$(1400 \div 5780) = 0.24$	$(4500 \times 0.24) = 1080$	1.09×10^3
$1400 \times 4500 = 6\,300\,000$	$(4500 \times 1400) \div 5780 = 1089.96$ or 1089.97 or 1090	1.0896×10^3
	6.3×10^6 (1+3ECF)	1.08×10^3
	2.4×10^{-1} (1+3ECF)	1.1×10^3
Mp1 incorrect but correct conversion to standard form		
	MP1 correct ÷ wrong denominator with correct standard form(1+3ECF)	

Question number	Answer	Additional guidance	Mark
4(b)(i)	An explanation that includes the following points: - chemoreceptors detect {lower pH of blood / increased blood CO₂ concentration}/ baroreceptors detect change in blood pressure (1) (electrical) {impulses sent to medulla (oblongata)/cardiovascular control centre (CVC)} / chemoreceptors are located in medulla oblongata (1) - impulses travel {from the medulla oblongata / via sympathetic nervous system} to the SAN /more impulses sent to {AVN/ bundle of His / Purkyne fibres} (1) (causing SAN) to increase frequency of depolarisation causing an increase in {heart rate / systole} (1)	Do not accept reference to signals / messages increased adrenaline production adrenaline which circulates in the blood mp3 adrenaline binds to receptors in {heart muscle / SAN} mp4 (causing SAN) to increase frequency of depolarisation causing an increase in {heart rate / systole} chemoreceptors in medulla oblongata detect increase in blood CO₂ = 2 marks	Expert (4)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	An answer that includes two of the following points: - fast twitch muscles fatigue quickly / are for short distance (sprinting) / cheetah cannot get oxygen fast enough for aerobic respiration (1) (cells / muscles) respire by anaerobic respiration (1) (therefore) build up of {lactic acid / oxygen debt} (from anaerobic respiration) (1)	the sprint requires large amounts of ATP (for muscle contraction) accept lactate accept release of heat so body temperature goes up rapidly / homeostatic mechanisms cannot occur quick enough	Expert (2)